Generating WikiArt Paintings with a DCGAN

ECE 285 HW2 `https://github.com/Tobudase/ece285-hw2-wikiart-gan`

Abstract

We compare three GAN configurations for 64×64 WikiArt generation. The wider DCGAN achieves the lowest validation FID (67.62), held-out test FID of 67.58, and Inception Score of 4.402. Generated paintings show clearer coarse structure, but fine detail remains limited.

1 Data and implementation

The assigned WikiArt archive [1] contains 81,444 images. We apply EXIF orientation, convert to RGB, center crop and resize bicubically to 64×64. Deduplication by exact processed pixels removes 747 images, leaving 80,697 images across 27 styles. A seeded content hash yields 64,615 training, 7,965 validation and 8,117 test images. Training ignores style labels; exact processed duplicates cannot cross splits, though near duplicates and shared artists may remain.

Our DCGAN [2] maps Gaussian noise through five 4×4 transposed convolutions ($512, 256, 128, 64, 3$ channels at width 64), with batch normalization/ReLU in hidden blocks and output Tanh. The discriminator uses strided convolutions ($64, 128, 256, 512$ channels), LeakyReLU(0.2), batch normalization except in the first block, and a final logit. Inputs use $[-1, 1]$; convolution weights use $\mathcal{N}(0, 0.02^2)$. Each step alternates a real-plus-fake BCE discriminator update and a non-saturating generator update; fake images are detached for the former.

Table 1: Configuration of the selected wider DCGAN.

Setting	Value	Setting	Value
Epochs / batch size	40 / 128	Latent / base width	100 / 64
Adam learning rate	2×10^{-4}	Adam (β_1, β_2)	$(0.5, 0.999)$
Training seed	285	Eval real / latent seeds	286 / 1286
Generator parameters	3,576,704	Discriminator parameters	2,765,568

Both new models train from scratch on one A100 80 GB (PyTorch 2.4.1, CUDA 12.1), with 504 full batches per epoch. Training takes 6.44 min for the wider DCGAN and 10.76 min for the variant, which uses nearest-neighbor resize plus convolution, spectral normalization on discriminator convolutions, and hinge losses.

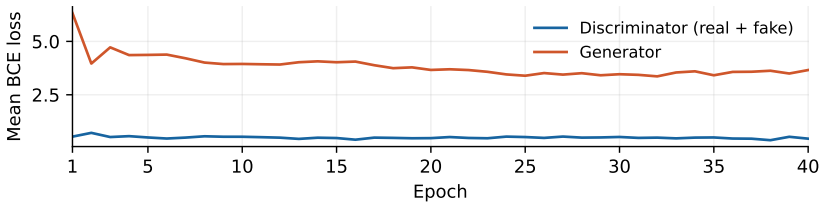


Figure 1: Selected model's epoch-mean training losses. The discriminator curve sums real and fake BCE.

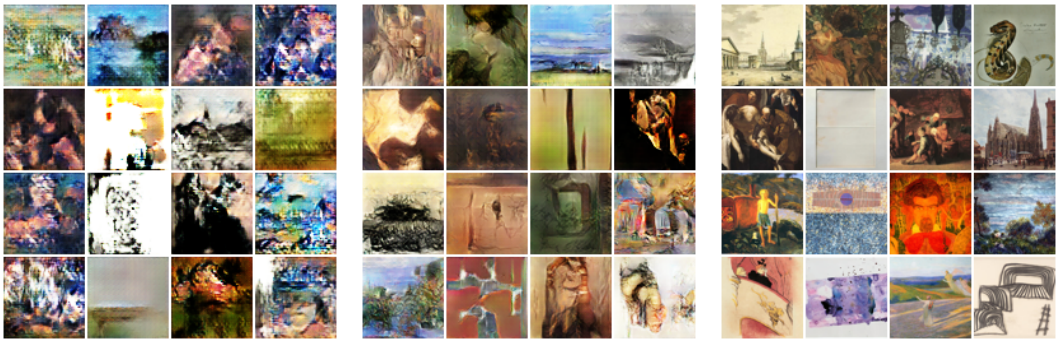
2 Evaluation and results

For each evaluation, we generate 5,000 images with noise seed 1286 and sample 5,000 real images without replacement (seed 286). Both use 8-bit RGB. Ignite computes the scores [3]. FID-2048 uses `pytorch-fid` pool3 features [4]; FID-1000 uses ImageNet probabilities. Inception inputs are resized bilinearly to 299×299. IS is one aggregate, with no split-based deviation. Reusing generated samples across splits explains matching IS values: IS depends only on generated images. All final checkpoints are compared on validation; the wider DCGAN is selected by lowest validation FID and then evaluated on test.

Table 2: Final checkpoints. Lower FID-2048 and higher IS are preferable within this protocol.

Model (width / epochs)	Val. FID	Val. IS	Test FID	Test IS
Original DCGAN (32 / 20)	112.81	3.212	111.84	3.213
Wider DCGAN (64 / 40)	**67.62**	4.402	**67.58**	**4.402**
Resize + SN + hinge (64 / 40)	75.77	4.713	—	—

The selected model lowers test FID by 39.6% relative to the original. Its test FID-1000 is 0.0750; real-image test IS is 7.522. Independent recomputation from saved features agrees with the metrics. Validation runs share CUDA; the original test result was computed on MPS with the same data and sampling protocol.



(a) Original DCGAN (b) Wider DCGAN (c) Real test images

Figure 2: First 16 seeded samples per group. Both generators use the same latent sampling protocol.

3 Discussion

The wider DCGAN shows more coherent landscape layers and some recognizable figure outlines, but textures and small objects remain distorted. The resize/SN variant has higher validation IS yet worse FID than the wider DCGAN. Late generator losses fluctuate around a plateau while discriminator loss remains low; these curves do not establish convergence. Width and training duration change together relative to the original, so the gain cannot be attributed to either alone. One training seed, low resolution, and ImageNet features limit conclusions about variability and artistic fidelity.

References

[1] Steubk. *WikiArt*, Kaggle dataset, version 1.

[2] A. Radford, L. Metz, and S. Chintala. Unsupervised representation learning with deep convolutional generative adversarial networks. ICLR, 2016.

[3] PyTorch-Ignite. GAN evaluation using FID and IS.

[4] M. Heusel et al. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. NeurIPS, 2017. Features: `pytorch-fid`.